

- 4 (a) State the **two** assumptions of the kinetic theory of gases that relate to collisions of particles with each other and with the walls of the container of the gas.

1

2

[2]

- (b) A single particle of an ideal gas has mass m . The particle is moving at speed u in a cubic box of side length L , as shown in Figure 4.1.

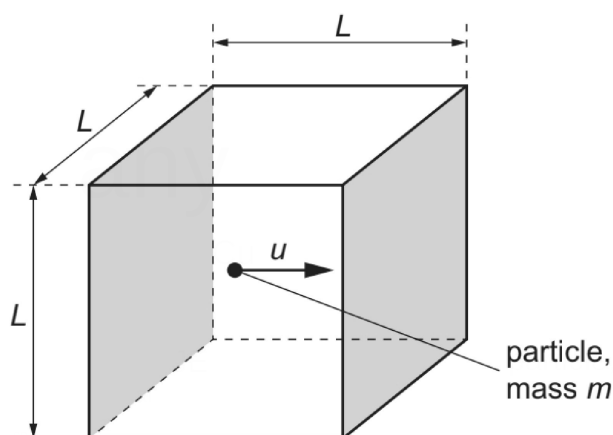


Figure 4.1

The particle is moving in a direction that is perpendicular to the two shaded faces, continually rebounding from one face and then the other.

Determine an expression, in terms of some or all of m , u and L , for:

- (i) the magnitude of the change in momentum of the particle during each collision with a shaded face

change in momentum = [1]

- (ii) the time taken between consecutive collisions of the particle with **one** of the shaded faces

time = [1]

- (iii) the mean force exerted on the particle by one of the shaded faces

mean force = [1]

- (iv) the mean pressure p exerted over the area of each of the shaded faces by the particle.
Explain your reasoning.

$p =$ [2]

- (c) An ideal gas consists of N particles moving in many different directions with a mean-square speed of $\langle c^2 \rangle$.

Explain how the equation in **4(b)(iv)** is modified to arrive at the more general equation

$$pV = \frac{1}{3} Nm \langle c^2 \rangle$$

where V is the volume of the gas.

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..... [3]

[Total: 10]